CPE 231 – Algorithms (1/69)

Homework IV

1. (15 points) Consider the following ranking matrix:

	Ann	Lea	Sue
Bob	2, 1	1, 2	3, 1
Jim	1, 2	2, 1	3, 2
Tom	1, 3	2, 3	3, 3

   (a) Find a stable marriage matching by applying the Gale-Shapley algorithm in its men proposing version. In each iteration, indicate the proposing person and the person rejecting/accepting the proposal, and the matching pairs.

   (b) Repeat (a) for the women proposing version.

   (c) Are the solutions in (a) and (b) different? Explain.

2. (15 points) Given the pattern $P =$ `BAOBAB` and $\Sigma = \{$`A,B,C,…, _` $\}$ (Capital english letters and space character), and the text $T =$ `BESS_KNEW_ABOUT_BAOBABS`

   (a) Apply the brute-force algorithm to the text. Determine the number of pattern shifts and the number comparisons made in each alignment. Start the comparison in each alignment from right to left.

   (b) Determine the shift table for Horspool's algorithm.

   (c) Apply Horspool's algorithm to the text and show how the pattern is aligned with the text and shifted in each step. Determine the number of pattern shifts and the number comparisons made in each alignment. What is the number of comparisons made until successful search ?